

Study and Empirical Analysis of Sentiment Analy...

By: Monish Gupta

As of: May 24, 2022 12:03:14 PM
4,203 words - 45 matches - 38 sources

Similarity Index

12%

Mode: Similarity Report ▼

paper text:

Chapter 13 Study and Empirical Analysis of Sentiment Analysis Approaches Monish Gupta¹, Sumedh Hambarde², Devika Verma³, Vivek Deshpande⁴ and Rakesh Ranjan⁵ ^{1,2,3,4} Vishwakarma Institute of Information Technology, India ⁵ Charles W Davidson College of Engineering, San Hoje State University, USA {¹monish.17u611, ²sumedh.17u570, ³devika.verma, ⁴vivek.deshpande}@viit.ac.in ⁵ rakesh.ranjan@sjsu.edu Abstract. Sentiment analysis is useful tool for social media and customer analysis allowing us to get summary of the views of large public behind a topic. This work conducts an empirical survey of different techniques for sentiment analysis; covering the implementation, advantages and limitations of each of these methods and conduct an experiment to find out which of these methods is best suited for sentiment analysis in today's scenario. Experimentation conducted on benchmark dataset and the results obtained highlight that supervised

learning algorithms like Support Vector Machines (SVM), Multinomial Logistic Regression (MLR) and Deep Learning based algorithms like Convolutional Neural Network

17

(CNN) and Bi-Directional Recursive Neural Network (RNN) show improved performances over the lexicon-based methods.

Keywords: Sentiment Analysis · Opinion Mining · Machine Learning · Lexicon Based Approach

23

· Deep Learning 1 Introduction "Data is the new oil", in the 21st century. But just like oil is an immensely valuable asset but of no value when unrefined, the huge amount of data when processed can yield valuable information which can help in various fields and applications. Considering the huge size of the data being generated every day, it is not possible to process all this data manually. Various techniques have been employed

to extract useful information from the data , one of

31

them being

sentiment analysis or opinion mining . Sentiment analysis is the process of finding the

7

sentiment, mostly positive, negative, neutral, given a portion of text. Apart from these categories, there are other categories in which the text 2 Gupta et al. can be classified into, e.g., happy, angry, sad, surprised, enthusiastic [16]. With the popularisation of Web 2.0, there has been an explosion in the amount of user-generated data on the internet. Sentiment analysis has been used in applications in a variety of fields. People express their opinions on popular social media sites as well as product review sites. Along with this, there are other sources of data like news articles related to all kinds of events like politics, entertainment, sports, finance, etc. Businesses can analyse the product reviews of their products posted online by customers and make suitable changes [11]. Analysis of product reviews posted on online shopping sites can help identify products which a customer is more likely to buy, boosting profits for the company, as well as increasing customer ease [31]. Sentiment analysis of reviews on travel and tourism websites [19] can help in recommending better hotels and destinations to tourists and boost the tourism industry, while sentiment analysis of news articles [26] can help in creation of customized news feeds sent to users. Movie review analysis has also been done to increase quality of movie recommendations [23,20]. Sentiment analysis has also been done to improve accuracy of predictions, like election-results [27] and movement

in stock -prices based on financial news [25,12]. In this work, we

24

review and implement different approaches for sentiment analysis; cover the working, advantages and limitations of each of these methods and conduct an experiment to find out which of these methods is best suited for sentiment analysis in today's scenario.

Paper is organized as follows: Section 2 covers the related work summarising popular sentiment analysis

11

techniques. This is followed by experiment design in section 3. We discuss

results in section 4 followed by conclusion and future **work in**

9

the last sections. Study and Empirical Analysis of Sentiment Analysis Approaches 3 2 Literature Survey Fig.1. Sentiment Analysis Architecture Sentiment analysis process usually includes three common

steps as shown in Figure 1. The first step is **pre-processing** of **the dataset**

21

, which involves removing stop words, null rows, html tags, url and converting text to lowercase. Once the

data is pre-processed, and features are extracted

15

sentiment classification is performed on it. Finally, the results of the analysis are evaluated by using confusion matrix to calculate precision, recall, accuracy and F1score. The various methods available for

sentiment classification can generally **be divided into**

37

lexicon-based methods, **machine-learning based** methods **and deep-learning based** methods, with **each** having **its own** advantages **and** **drawbacks**

5

2.1 Lexicon-based methods Lexicon-based methods

30

consist of creating a lexicon of sentiment words, and the sentiment scores associated with them. Sentiment words are those positive or negative words which are deemed to be important in calculating sentiment. [9] proposed an approach to predict sentiment

based on frequency of occurrences of **positive and negative words in**

25

a sentence; this method can be broadly described as “lexicon-based method”. Popular lexicons are SentiWordnet, Liu and Hu opinion lexicon, SentiWords [30]. SentiWordnet assigns to each sentiment word a positivity score, a negativity score and an objectivity score. 4 Gupta et al. SentiWords assigns to the words a score between -1 and 1 based on the positivity and the negativity. The principal advantage of the lexicon-based methods is that as they are unsupervised methods, they have no need for a training phase or a labelled dataset. But the disadvantage being that the contextual information is ignored, e.g., in the sentence “The product is cheap and easy-to-use, but it stopped working within 1 month”, there are more positive words (cheap, easy-to-use) than negative words (stopped), so probability of it being classified as positive is higher, though from the context we can see that it is a negative review. Certain work has been done to overcome this disadvantage of lexicon-based methods by combining it with other deep-learning based methods. [31] proposes a model for sentiment analysis by combining lexicon-based methods and deep-learning. Their method uses lexicon for enhancing sentiment features and using

deep learning , feature **for sentiment analysis** are extracted. **Neural networks and**

13

SentiWordnet lexicon were combined to perform sentiment analysis by [23]. 81 major non-English language lexicons were gathered by [5]. 2.2 Machine-learning based methods Several research highlight the recent shift towards statistical approach for sentiment analysis, relying on sufficiently large datasets for supervised training. A hybrid sentiment classification model of adaboosted decision tree and SVM was proposed in [21].

The preprocessed data **is given as input to the SVM and** adaboosted **decision tree** followed by **their outputs**

2

further

given as input to the decision tree

2

. The Naïve bayes and SVM machine learning models were trained on amazon dataset which contains reviews of various gadgets and electronics products like Camera, smart phones etc. and [11] reported an accuracy of 98.17% for Naïve bayes and 93.54% using SVM for sentiment analysis of Camera Reviews. The authors of [8] compared three sentiment analysis lexicon methods named SentiWordnet, TextBlob and W-WSD along with

the machine learning techniques, **Naïve bayes** and **SVM** . A study **and**

4

comparison between multinomial Naïve bayes, SVM, Maximum Entropy, Bernoulli

Naïve bayes, and Decision Trees was carried out **in**

4

[20]. [7] show a comparative study of

Naïve Bayes, SVM, Random Forest and

4

Study and Empirical Analysis of Sentiment Analysis Approaches 5 Maximum Entropy on the google reviews dataset. [17] proposed the use of Naïve bayes method for sentiment analysis and to tag them as positive or negative whereas [24] highlight SVM outperforms

compared to Naïve bayes and maximum entropy methods

8

. 2.3 Deep-Learning Based Methods Studies highlight that with smaller dataset although machine learning methods are preferable, provided availability of sufficiently large dataset, deep learning algorithms have outperformed other approaches to sentiment analysis. For sentiment analysis task, results obtained in [28] show multilayer perceptron outperforms over other algorithms like SVM and random forests. [12] proposed

the use of Bidirectional Long Short-Term Memory (Bi-LSTM

22

) units for extracting aspect from microblog and headline. A novel

sentiment analysis methodology based on the lexicon

18

and combining

attention-based Bidirectional Gated Recurrent Unit (BiGRU

18

) and CNN is proposed in [31]. Bi-LSTM model was proposed on the Militarylife board on PTT website data in [14] along

with the support of a self-developed military sentiment dictionary

6

[25] proposed the hybrid model combining recurrent neural network(RNN) with LSTM to train the News Archive Data from Thompson Reuters. Latest studies [1] show the implementation of various deep learning algorithms and use of SentiCNN model[10] for solving several challenges of sentiment analysis.

2.4 Datasets

Datasets used for sentiment analysis differ based on the field of application. Most of the research involves using publicly available data from online product review websites. The Movie Review Dataset issued by Stanford consisting of 50,000 IMDb movie reviews which are labelled positive or negative [23]. Work done by [19] consolidates reviews from popular hotel review website TripAdvisor. [31] uses product reviews from famous Chinese shopping website dangdang.com, while lot of studies make use Amazon reviews dataset, BBC news dataset [26], tweets from Twitter [21,8,32] for sentiment- 6 Gupta et al. analysis. The benefits of using a twitter dataset are more information and sentiments expressed in lesser words, availability of a vast dataset, and possibility of real-time analysis.

3 Experiment Methodology

For empirical analysis of different sentiment analysis approaches, we setup an experiment over three different datasets over a variety of methods described in Figure 2.

Fig.2. Sentiment Classification Methods Used for Experimentation

Considering the wide variety of applications of sentiment analysis, we selected following datasets to be used in the experiment;

1. Large Movie Review Dataset [15]: This is a dataset provided by the Stanford University and consists of 25k movie reviews for training testing separately each. It consists of two classes (positive = 1 and negative = 0). Out of the 25,000 reviews each provided for training and testing, 12,500 reviews are positive and the remaining 12,500 reviews are negative.
2. Sentiment 140 Dataset [2]: This dataset consists of 1.6 million tweets extracted from twitter and consists of two classes (0 = negative and 4 = positive)
3. Amazon Baby Dataset: This dataset consists of reviews from amazon and has 3 columns (name, review and rating). Here name field contains the subject of the reviews, review field contains the text data

Study and Empirical Analysis of Sentiment Analysis Approaches

7 and ratings field consists of score of review between 1 to 5, 5 being highest rated.

3.1 Pre-processing

To extract the features from text which are not explicitly available like in structured data, we need to clean data to remove unwanted characters and perform pre-processing on all the corpus.

a Pre-processing on Large Movie Reviews Dataset

1. Removing HTML tags: HTML tags contained in reviews field are removed.
2. Lowercase: Converted the reviews to lower case.
3. Stopwords: Removed the stopwords contained in reviews.

b Pre-processing on Sentiment140 Dataset

2. Trimming the dataset: Trimmed the dataset to contain 12,500 positive and negative reviews each.
3. Removing HTML decoding: Removing HTML decoding like & and ' etc.
4. Removing @username: Removed the username contained in the tweets.
5. Removing URL links: Removed URL links contained in the tweets.
6. Hashtags: Hashtags might contain important information which is necessary to understand the sentiment so we just removed the hashtag symbol and kept the text under it in our reviews.
7. Lowercase: Converted the tweets to lowercase.
8. Stopwords: Removed the stopwords contained in the tweets.

a Pre-processing on amazon-baby Dataset

1. Remove null rows: Remove the rows containing empty fields.
2. Add sentiment field based on ratings: We assigned the review a negative sentiment if it has rating 1 or 2 and assigned the review a positive sentiment if it has rating 4 or 5. We dropped the reviews 8 Gupta et al. containing the rating as 3 since we only wanted two sentiment classes, positive and negative.
3. Trimming the dataset: We trimmed the dataset to contain 12,500 positive and negative reviews separately each.
4. Combined the name and review field: Since both the columns are important in sentiment classification, we combined them into one column to pass it to our model.
5. Lowercase: The reviews are converted to lowercase.
6. Stopwords: The stopwords are removed from the reviews.

Most of the pre-processing steps were implemented using source python library re [29] and NLTK [4]. Additionally, we even performed vectorization and train test split on all the 3 datasets. In case of machine learning for vectorization we used the tf-idf

vectorizer and the values of their attributes are as follows. (max_features = 5000, min_df = 5 and max_df = 0.7). In case of deep learning, we first padded the reviews and then for vectorization we used word embeddings provided by Keras [6]. Then we perform a 70:30 train test split of the dataset. 3.2 Methods Lexicon-based In this method, we used the SentiWordnet lexicon [3], which is the most commonly used lexicon for lexicon-based sentiment analysis. A lexicon positivity, negativity and objectivity scores is assigned for every synset within the Wordnet. If the positivity score is greater than the negativity score, then the predicted sentiment is "positive" and vice versa. The labelled and predicted sentiments are then compared to calculate

accuracy, precision, recall and f1-score for the method

16

. Machine Learning

We train the sentiment classification model using six different machine learning algorithms

35

i.e., SVM,

Naïve Bayes, Decision Trees , Multinomial Naïve bayes, KNN and

38

Multinomial Logistic Regression. All the six models have been implemented using the open-source python sklearn library [18]. The attributes of each of these methods are described below. Study and Empirical Analysis of Sentiment Analysis Approaches 9 1. SVM: Regularization parameter is set to 1, kernel = linear and gamma = 'auto'. 2. Naïve Bayes: Default attributes provided by sklearn. 3. Multinomial Logistic Regression: Attribute multi class = 'multinomial' and solver = 'newton-cg'. 4. Multinomial Naïve Bayes: Default attributes provided by sklearn. 5. KNN: Attribute n_neighbours = 7. 6. Decision Trees: Default attributes provided by sklearn. Deep Learning For deep learning we implement Bi-directional Recurrent Neural Network [22] and 1D Convolutional Neural Network [13]. Both of these models have been implemented by using the Keras library. 3.3 Evaluation metrics The performances of the different sentiment classification techniques are measured using following metrics:

True Positive (TP) : Positive text **classified as positive. False Positive (FP)** : Negative text **classified as positive** . True **Negative** (TN): Negative text **classified as negative** . False **Negative** (FN): Positive text **classified as Negative. Accuracy**

=

$TP + TN$ * 100 $TP + TN + FP + FN$ Precision: **The ratio of** true positives **to the**

19

total reviews classified as positive. $Precision = TP * 100 / (TP + FP)$ Recall:

The ratio of True Positives to total actual **positives**

32

. 10 Gupta et al.

$Recall = TP * 100 / (TP + FN)$ **F1: The weighted average of precision and recall. $F1 = 2 * precision * recall / (precision + recall)$**

14

4 Results 4.1 Results of Lexicon Based Methods The results of lexicon-based models

are shown in table 1. The average F-score

3

achieved using lexicon-based method is 62.33%. Dataset SentiWordnet Method Accuracy Precision Recall F1-Score Large Movie Reviews Dataset 64.54% 66.5% 66.5% 66.5% Sentiment 140 Dataset 58.46% 59.5% 58.5% 57% Amazon Baby Dataset 63.31% 63.5% 62% 63.5% Table 1. Results of SentiWordnet 4.2 Results of Machine Learning Based

Methods The results of training models using different machine learning algorithms are shown in tables 2 to 7. The average F1 score using SVM is 84%, using Naïve Bayes is 74.50%, using Multinomial Naïve Bayes is 82%, using Decision Trees is 71%, using Multinomial Logistic Regression is 84.33% and using K Nearest Neighbours is 71.16%. The result highlight that we achieved the best performance using the Multinomial Logistic Regression algorithm. Dataset Support Vector Machines Study and Empirical Analysis of Sentiment Analysis Approaches 11 Accuracy Precision Recall F1-Score Large Movie Reviews Dataset 87.94% 88% 88% 88% Sentiment 140 Dataset 72.92% 73% 73% 73% Amazon Baby Dataset 90.56% 90.5% 91% 91% Table 2. Results of Support Vector Machines Dataset Naïve Bayes Accuracy Precision Recall F1-Score Large Movie Reviews Dataset 80.03% 80% 80% 80% Sentiment 140 Dataset 65.2% 65.5% 65% 65% Amazon Baby Dataset 78.66% 79.5% 78.5% 78.5% Table 3. Results of Naïve Bayes Dataset Mutinomial Naïve Bayes Accuracy Precision Recall F1-Score Large Movie Reviews Dataset 86.04% 86% 86.5% 86% 12 Gupta et al. Sentiment 140 Dataset 73.17% 73.5% 73.5% 73% Amazon Baby Dataset 87.18% 87% 87.5% 87% Table 4. Results of Multinomial Naïve Bayes Dataset Decision Trees Accuracy Precision Recall F1-Score Large Movie Reviews Dataset 71.12% 71% 70% 71% Sentiment 140 Dataset 67.02% 67% 66.5% 67% Amazon Baby Dataset 74.93% 75% 75.5% 75% Table 5. Results of Decision Trees Dataset Multinomial Logistic Regression Accuracy Precision Recall F1-Score Large Movie Reviews Dataset 88.69% 88.5% 88% 88.5% Study and Empirical Analysis of Sentiment Analysis Approaches 13 Sentiment 140 Dataset 73.36% 73.5% 72.5% 73.5% Amazon Baby Dataset 90.84% 90.5% 90.5% 91% Table 6. Results of Multinomial Logistic Regression Dataset K-Nearest Neighbours Accuracy Precision Recall F1-Score Large Movie Reviews Dataset 75.6% 76.5% 76% 75.5% Sentiment 140 Dataset 57.3% 57.5% 57.5% 57% Amazon Baby Dataset 81.22% 82% 81% 81% Table 7. Results of K Nearest Neighbours 4.3 Results of Deep Learning Based Methods

We have used accuracy as an evaluation metric

27

for deep learning

models. The results obtained **are shown in** the **table** 8. **The**

9

average accuracy for Bi-directional Recurrent Neural Network is 83.03% and the average accuracy for 1D Convolutional Neural Network is 80.32%. The results of deep learning methods highlight that Bi-directional Recurrent Neural Network provides slightly better results than 1D Convolutional Networks. Bi-Directional RNN 1D- CNN Large Movie Reviews Dataset 88.42% 83.7% Sentiment 140 70.21% 69.67% 14 Gupta et al. Dataset Amazon Baby Dataset 90.46% 87.61% Table 8. Results of Deep Learning Methods The results obtained from the above experiment show that lexicon- based method does not give the best results mostly due to the inability of the lexicon-based method to integrate the contextual information into the process; treating the sentence

as a bag-of-words and ignoring **the word** order **of the**

10

sentences; apart from that negation handling is a feature that

needs to be dealt with **to** improve **the accuracy of the model**

12

. Lexicon-based method can be integrated with deep learning methods to best handle these problems. Amongst machine

learning algorithms, Support Vector Machines and **Multinomial Logistic Regression**

33

gave higher results compared to other algorithms we experimented with. For Deep Learning Approach, both algorithms have shown similar performances with the Bi-directional RNN giving slightly better results than 1D convnets. The precision value highlights the ability of the trained model not to classify a negative review as positive. The recall value is

the ability of the trained model **to find all the positive samples** . From **the**

3

obtained results it is observed that the difference between precision and recall values is less which shows that equal amount of dataset has been classified as false positive by the algorithm as it has classified false negatives. This is valid as the dataset had approximately equal sets of positive and negative reviews. Study and Empirical Analysis of Sentiment Analysis Approaches 15 5 Conclusion This paper compares the best models under the three main types of sentiment analysis methods which

are lexicon-based, machine learning and deep learning methods

26

. It was found that

machine learning and deep learning methods provide **better** results than **the**

28

lexicon-based methods, provided the availability of relevant dataset. Through the experiment conducted, we identify that SVM and Multinomial logistic regression provided better results than the other machine learning methods mentioned. For deep learning approach, both algorithms show similar performances with bi-directional RNN giving slightly better results than 1D-convnets. 6 Future Work Apart from the huge amount of textual data, more and more audio- video data is being produced every day with the rise of social media companies like Instagram, Facebook, etc.

Most of the current research is focused on

34

sentiment analysis of textual data; however, work needs to be furthered

in the fields of image **sentiment analysis and** video **sentiment analysis**

13

. Sarcasm detection is another challenge of sentiment analysis in which further work needs to be done; as people convey negative emotions using positive words, there is high probability of the review being misclassified as positive instead of negative. While

there has been a **considerable amount of research done in sentiment analysis**

20

in non-English languages, the major challenges which face us are the lack of availability of quality lexicons for these languages, and the usage of mixed language. Further work can be done to introduce a standard cross-lingual approach framework and develop quality lexicons for other languages.

Most of the work done **so far focuses on**

29

classifying the text into positive or negative

36

. But for specific projects, a more detailed categorization of the text may be needed, e.g., angry, disappointed, happy, enthusiastic, etc. Work needs to be done to make the classifications more detailed and accurate. 16 9. Gupta et al. References 1. Ain, Q.T., Ali, M., Riaz, A., Noureen, A., Kamran, M., Hayat, B., Rehman, A.: Sentiment analysis using deep learning techniques: A review. International Journal of Advanced Computer Science and Applications 8(6) (2017). <https://doi.org/10.14569/IJACSA.2017.080657>, <http://dx.doi.org/10.14569/IJACSA.2017.080657> 2. Alec Go, R.B., Huang, L.: Twitter sentiment classification using distant supervision. CS224N Project Report, Stanford, 1 (2009) 3. Baccianella, S., Esuli, A., Sebastiani, F.: Sentiwordnet 3.0: An enhanced lexical resource for sentiment analysis and opinion mining. In: Calzolari, N., Choukri, K., Maegaard, B., Mariani, J., Odijk, J., Piperidis, S., Rosner, M., Tapias, D. (eds.) LREC. European Language Resources Association (2010), <http://nmis.isti.cnr.it/sebastiani/Publications/LREC10.pdf> 4. Bird, S., Klein, E., Loper, E.: Natural language processing with Python: analyzing text with the natural language toolkit." O'Reilly Media, Inc." (2009) 5. Chen, Y., Skiena, S.: Building sentiment lexicons for all major languages. In: ACL (2014) 6. Chollet, F., et al.: Keras. <https://keras.io> (2015) 7. Gomathi, D.S.: Sentiment analysis of google reviews of a college (2019) 8. Hasan, A., Moin, S., Karim, A., Band, S.: Machine learning-based sentimental analysis for twitter accounts. Mathematical and Computational Applications 23, 11 (02 2018). <https://doi.org/10.3390/mca23010011> 9. Hu, M., Liu, B.: Mining and summarizing customer reviews. In: Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. p. 168–177. KDD '04, Association for Computing Machinery, New York, NY, USA (2004). <https://doi.org/10.1145/1014052.1014073>, <https://doi.org/10.1145/1014052.1014073> 10. Huang, M., Xie, H., Rao, Y., Liu, Y., Poon, L.K.M., Wang, F.L.: Lexicon- based sentiment convolutional neural networks for online review 12 Gupta et al. analysis. IEEE Transactions on Affective Computing pp. 1–1 (2020). <https://doi.org/10.1109/TAFFC.2020.2997769> 11. Jagdale, R., Shirsat, V., Deshmukh, S.: Sentiment Analysis on Product Reviews Using Machine Learning Techniques: Proceeding of CISC 2017, pp. 639–647 (01 2019). https://doi.org/10.1007/978-981-13-0617-4_61 12. Jangid, H., Singhal, S., Shah, R.R., Zimmermann, R.: Aspect-based financial sentiment analysis using deep learning. pp. 1961–1966 (04 2018). <https://doi.org/10.1145/3184558.3191827> 13. Kiranyaz, S., Avci, O., Abdeljaber, O., Ince, T., Gabbouj, M., Inman, D.J.: 1d convolutional neural networks and applications: A survey. Mechanical Systems and Signal Processing 151, 107398 (2021). <https://doi.org/https://doi.org/10.1016/j.ymssp.2020.107398>, Study and Empirical Analysis of Sentiment Analysis Approaches 17 <https://www.sciencedirect.com/science/article/pii/S0888327020307846> 14. Liang-Chu Chen, C.M.L., Chen, M.Y.: Exploration of social media for sentiment analysis using deep learning. Soft Computing 24, 8187–8197 (2020). <https://doi.org/10.1007/s00500-019-04402-8> 15. Maas, A.L., Daly, R.E., Pham, P.T., Huang, D., Ng, A.Y., Potts, C.: Learning word vectors for sentiment analysis. In: Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language

Technologies. pp. 142–150. Association for Computational Linguistics, Portland, Oregon, USA (June 2011), <http://www.aclweb.org/anthology/P11-1015> 16. Nandwani, P., Verma, R.: A review on sentiment analysis and emotion detection from text. Soc. Netw. Anal. Min. 11 (august 2021). <https://doi.org/10.1007/s13278-021-00776-6> 17. Omkar Borade, Kaushik Gosavi, A.G., Shinde, A.: "sentiment analysis of college reviews". vol. 5, pp. 319–322 (04 2017), <http://www.ijedr.org/papers/IJEDR1702054.pdf> 18. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, E.: Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830 (2011) 19. Prameswari, P., Zulkarnain, Surjandari, I., Laoh, E.: Mining online reviews in indonesia's priority tourist destinations using sentiment analysis and text summarization approach. In: 2017 IEEE 8th International Conference on Awareness Science and Technology (iCAST). pp. 121–126 (2017). <https://doi.org/10.1109/ICAWSST.2017.8256429> 20. Rahman, A., Hossen, M.S.: Sentiment analysis on movie review data using machine learning approach. In: 2019 International Conference on Bangla Speech and Language Processing (ICBSLP). pp. 1–4 (2019). <https://doi.org/10.1109/ICBSLP47725.2019.201470> 21. Rathi, M., Malik, A., Varshney, D., Sharma, R., Mendiratta, S.: Sentiment analysis of tweets using machine learning approach. In: 2018 Eleventh International Conference on Contemporary Computing (IC3). pp. 1–3 (2018). <https://doi.org/10.1109/IC3.2018.8530517> 22. Schuster, M., Paliwal, K.: Bidirectional recurrent neural networks. IEEE Transactions on Signal Processing 45(11), 2673–2681 (1997). <https://doi.org/10.1109/78.650093> 23. Shaukat, Z., Zulfiqar, A.A., Xiao, C., Azeem, M., Mahmood, T.: Sentiment analysis on imdb using lexicon and neural networks 2 (2020). <https://doi.org/10.1007/s42452-019-1926-x> 24. Shivaprasad, T.K., Shetty, J.: Sentiment analysis of product reviews: A review. In: 2017 International Conference on Inventive Communication and Computational Technologies (ICICCT). pp. 298–301 (2017). <https://doi.org/10.1109/ICICCT.2017.7975207> 25. Gupta et al. 26. Souma, W., Vodenska, I., Aoyama, H.: Enhanced news sentiment analysis using deep learning methods. Journal of Computational Social Science 2 (2019). <https://doi.org/10.1007/s42001-019-00035-x> 27. Taj, S., Meghji, A., Shaikh, B.: Sentiment analysis of news articles: A lexicon based approach (02 2019). <https://doi.org/10.1109/ICOMET.2019.8673428> 28. Tjong Kim Sang, E., Bos, J.: Predicting the 2011 Dutch senate election results with Twitter. In: Proceedings of the Workshop on Semantic Analysis in Social Media. pp. 53–60. Association for Computational Linguistics, Avignon, France (Apr 2012), <https://www.aclweb.org/anthology/W12-0607> 29. Valencia, F., Gómez-Espinosa, A., Valdés-Aguirre, B.: Price movement prediction of cryptocurrencies using sentiment analysis and machine learning. Entropy 21(6), 589 (Jun 2019). <https://doi.org/10.3390/e21060589>, <http://dx.doi.org/10.3390/e21060589> 30. Warriner, A., Kuperman, V., Brysbaert, M.: Norms of valence, arousal, and dominance for 13,915 english lemmas. Behavior research methods 45 (02 2013). <https://doi.org/10.3758/s13428-012-0314-x> 31. Yang, L., Li, Y., Wang, J., Sherratt, R.S.: Sentiment analysis for e-commerce product reviews in chinese based on sentiment lexicon and deep learning. IEEE Access 8, 23522–23530 (2020). <https://doi.org/10.1109/ACCESS.2020.2969854> 32. Zabha, N., Ayop, Z., Anawar, S., Erman, H., Zainal, Z.: Developing cross-lingual sentiment analysis of malay twitter data using lexicon-based approach. International Journal of Advanced Computer Science and Applications 10 (01 2019). <https://doi.org/10.14569/IJACSA.2019.010014>

sources:

- 1 21 words / 1% - Crossref
["Intelligent Systems Design and Applications", Springer Science and Business Media LLC, 2018](#)

- 2 21 words / 1% - Crossref
[Megha Rath, Aditya Malik, Daksh Varshney, Rachita Sharma, Sarthak Mendiratta. "Sentiment Analysis of Tweets Using Machine Learning Approach", 2018 Eleventh International Conference on Contemporary Computing \(IC3\), 2018](#)

- 3 19 words / 1% - Crossref
["Chinese Computational Linguistics and Natural Language Processing Based on Naturally Annotated Big Data", Springer Science and Business Media LLC, 2013](#)

- 4 19 words / 1% - Crossref
[Pansy Nandwani, Rupali Verma. "A review on sentiment analysis and emotion detection from text", Social Network Analysis and Mining, 2021](#)

- 5 14 words / < 1% match - Internet from 19-Dec-2021 12:00AM
[link.springer.com](#)

- 6 10 words / < 1% match - Internet from 17-Nov-2021 12:00AM
[link.springer.com](#)

- 7 9 words / < 1% match - Internet from 25-Mar-2019 12:00AM
[link.springer.com](#)

- 8 8 words / < 1% match - Internet from 30-Oct-2020 12:00AM
[link.springer.com](#)

- 9 17 words / < 1% match - Internet from 27-Jun-2018 12:00AM
[aclweb.org](#)

- 10 9 words / < 1% match - Internet from 16-Mar-2016 12:00AM
[aclweb.org](#)

-
- 11 12 words / < 1% match - Internet from 16-Dec-2021 12:00AM
[ebin.pub](#)
-
- 12 9 words / < 1% match - Internet from 03-Mar-2022 12:00AM
[ebin.pub](#)
-
- 13 17 words / < 1% match - Internet from 23-May-2022 12:00AM
[ikee.lib.auth.gr](#)
-
- 14 16 words / < 1% match - Internet from 01-Jan-2022 12:00AM
[centaur.reading.ac.uk](#)
-
- 15 8 words / < 1% match - Internet from 03-Mar-2021 12:00AM
[dokumen.pub](#)
-
- 16 8 words / < 1% match - Internet from 20-Nov-2021 12:00AM
[dokumen.pub](#)
-
- 17 14 words / < 1% match - Crossref
[Ilyos Rabbimov, Sami Kobilov, Iosif Mporas. "Chapter 53 Opinion Classification via Word and Emoji Embedding Models with LSTM", Springer Science and Business Media LLC, 2021](#)
-
- 18 13 words / < 1% match - Crossref
[Li Yang, Ying Li, Jin Wang, R. Simon Sherratt. "Sentiment Analysis for E-Commerce Product Reviews in Chinese Based on Sentiment Lexicon and Deep Learning", IEEE Access, 2020](#)
-
- 19 11 words / < 1% match - Crossref
["Artificial Intelligence for Cyber Security: Methods, Issues and Possible Horizons or Opportunities", Springer Science and Business Media LLC, 2021](#)
-
- 20 11 words / < 1% match - Crossref

[Kaushik Dhola, Mann Saradva. "A Comparative Evaluation of Traditional Machine Learning and Deep Learning Classification Techniques for Sentiment Analysis", 2021 11th International Conference on Cloud Computing, Data Science & Engineering \(Confluence\), 2021](#)

21 11 words / < 1% match - Crossref
[Prabhu Ram Nagarajan, Meeradevi. T, Shri Ram A.M Hari, Kavippriya. K, Madhumitha G. "Certain Investigation On Cause Analysis Of Accuracy Metrics In Sentimental Analysis On News Articles", 2021 5th International Conference on Electronics, Communication and Aerospace Technology \(ICECA\), 2021](#)

22 10 words / < 1% match - Internet from 24-Dec-2021 12:00AM
[Repositorium.uminho.pt](#)

23 10 words / < 1% match - Internet from 05-Dec-2021 12:00AM
[www.jatit.org](#)

24 9 words / < 1% match - Crossref
["Computational Science and Technology", Springer Science and Business Media LLC, 2021](#)

25 9 words / < 1% match - Crossref
[Lecture Notes in Computer Science, 2014.](#)

26 9 words / < 1% match - Internet from 13-Nov-2021 12:00AM
[arxiv.org](#)

27 8 words / < 1% match - Crossref
["Data Intelligence and Cognitive Informatics", Springer Science and Business Media LLC, 2022](#)

28 8 words / < 1% match - Internet from 04-Mar-2022 12:00AM
[Dergipark.Org.Tr](#)

29 8 words / < 1% match - ProQuest
[Rolston, Leanne. "Dialogical Signals of Stance Taking in Spontaneous Conversation.", University of Washington, 2021](#)

30

8 words / < 1% match - Internet from 08-Feb-2022 12:00AM
deepai.org

31

8 words / < 1% match - Internet from 22-Feb-2022 12:00AM
etheses.whiterose.ac.uk

32

8 words / < 1% match - Internet from 01-Mar-2021 12:00AM
journal.iberamia.org

33

8 words / < 1% match - Internet from 23-Nov-2019 12:00AM
jyx.jyu.fi

34

8 words / < 1% match - Internet from 18-Mar-2022 12:00AM
medinform.jmir.org

35

7 words / < 1% match - Crossref
["Emerging Technologies in Data Mining and Information Security", Springer Science and Business Media LLC, 2021](#)

36

7 words / < 1% match - Crossref
[Communications in Computer and Information Science, 2014.](#)

37

6 words / < 1% match - Crossref
["2020 International Conference on Applications and Techniques in Cyber Intelligence", Springer Science and Business Media LLC, 2021](#)

38

6 words / < 1% match - Crossref
["Computer Networks and Inventive Communication Technologies", Springer Science and Business Media LLC, 2022](#)